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Multi-Currency Portfolio Optimization with General Risk Measures

Master Thesis

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Acknowledgments

Abstract

Multi-currency portfolio construction requires managing asset and currency exposures together, yet it remains unclear whether these decisions are best made simultaneously or sequentially as the investable universe grows. This thesis builds on the regularized multi-currency Expected Shortfall framework of Lucescu and Ulrych (2026), in which portfolios are formed by minimizing a risk objective subject to L^1 and L^2 penalties, with currency exposures either optimized jointly with assets or handled through a separate overlay. First we replicate the framework's core strategies, then extend it along two dimensions, the size of the investable universe, and the choice of objective function, implementing both Expected Shortfall and mean-variance formulations of the joint and overlay optimizers.

Across both objectives, the joint optimizer's out-of-sample performance deteriorates relative to the separate overlay as the universe expands. This motivates the central question of the thesis, is this deterioration a structural property of joint optimization, or is it modulated by the choice of objective function? To answer it, we develop a general regularized optimization framework that admits arbitrary objective functions within a single, common formulation. We validate the framework by showing that it exactly reproduces the existing Expected Shortfall and mean-variance optimizers, and then use it to measure how the rate of the joint optimizer's breakdown varies as dimensionality increases.

Sentence on results.

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Chapter 1

Introduction

1.1 Motivation: International Diversification, Currency Risk, and Tail Risk

1.2 Contributions

1.3 Thesis Structure

Chapter 2

Literature Review

Cite the following: International allocation and currency risk management, Currency overlay vs. joint optimization, Downside risk measures and Expected Shortfall, Instability of ES optimization and the role of regularization, Generalized objectives and decision-focused / end-to-end learning.

In this chapter, we review the relevant literature on multi-currency portfolio optimization. We discuss the contributions and methodologies in the field, highlighting how our work builds upon and extends existing research.

The core of our analysis builds on two recent studies that apply regularization techniques to international asset allocation. Burkhardt and Ulrych (2023) introduce a joint optimization approach within a mean-variance framework. By using L1 and L2 regularization, they show that jointly optimizing assets and currencies leads to better out-of-sample performance than the standard industry practice of using a separate currency overlay.

Extending this framework, Lucescu and Ulrych (2026) consider an Expected Shortfall optimization framework to better manage downside tail risk. To fix the numerical instability common in Expected Shortfall minimization, they apply similar regularization methods. Their findings reveal a dimensional trade-off. Joint optimization works best in small asset universes, but the separate currency overlay approach becomes more effective for larger universes due to its built-in dimensionality reduction.

Reproducing and comparing these two baseline frameworks serves as the starting point for the extensions proposed in this thesis.

Chapter 3

Optimization Framework

This chapter develops the optimization framework used throughout the thesis, building up from a model of multi-currency portfolio returns to the risk objectives, the regularization that stabilizes them, and the two formulations through which asset and currency decisions are combined. The derivations follow the methodology presented in [19] and [18], use simple returns, and assume that rebalancing occurs at the end of each investment period. We begin in Section 3.1 with a model-free representation of one-period portfolio returns for an investor who holds assets denominated in multiple currencies and can take currency forward positions for hedging, decomposing the unhedged return of a foreign asset into its local, currency, and cross-currency components, introducing the currency-mapping matrix that links assets to their implicit currency exposures, and expressing the hedged portfolio return in terms of net currency exposures.

With the return model in place, we formulate the optimization problem. Section 3.2 states the classical mean-variance criterion, and Section 3.3 introduces Expected Shortfall as a coherent downside-risk measure, together with the Rockafellar–Uryasev representation that casts its minimization as a tractable linear program. Both are presented as instances of a common mean–risk objective, so that the framework treats the risk measure as a swappable component rather than a fixed choice. Section **??** then introduces the L_1 and L_2 penalties and the leverage constraint used to stabilize the optimization and improve out-of-sample performance.

Finally, Sections **??** and **??** assemble these elements into the two approaches to multi-currency allocation studied throughout the thesis: the separate (currency overlay) formulation, in which asset and currency decisions are made sequentially, and the joint formulation, in which they are optimized simultaneously. The detailed derivations of the resulting quadratic programs are found in Appendix A and Appendix B.

3.1 Model-Free Multi-Currency Portfolio Returns

3.1.1 Unhedged Portfolio Returns

We consider an international investor who allocates their wealth across N assets denominated in up to $M + 1$ currencies. We fix a discrete time index $t \in \{0, 1, 2, \dots\}$ and one investment period from t to $t+1$. Let the investor's *domestic currency* be indexed by 0 and let the remaining M foreign currencies be indexed by $\mathcal{C} \coloneqq \{1, 2, \dots, M\}$.

Let $P_{i,t} > 0$ be the price of asset i at time t in its local currency $c_i \in \mathcal{C} \cup \{0\}$. Define the

corresponding *simple local-currency asset return* over $(t, t{+}1]$ as

$$R_{i,t+1} \coloneqq \frac{P_{i,t+1} - P_{i,t}}{P_{i,t}}.$$

Thus $P_{i,t+1} = P_{i,t}(1 + R_{i,t+1})$.

For each foreign currency $c \in \mathcal{C}$, let $S_{c,t} > 0$ denote the exchange rate at time t, quoted as units of domestic currency per one unit of foreign currency c. The associated *simple currency return* over $(t, t{+}1]$ is

$$e_{c,t+1} \coloneqq \frac{S_{c,t+1} - S_{c,t}}{S_{c,t}}.$$

Equivalently, $S_{c,t+1} = S_{c,t}(1+e_{c,t+1})$. For the domestic currency, we set $S_{0,t} \equiv 1$ and $e_{0,t+1} \equiv 0$.

The domestic-currency value of holding one unit of asset i at time t is $P_{i,t}\, S_{c_i,t}$. Therefore, the *unhedged domestic-currency simple return* of asset i is

$$\begin{aligned} R^u_{i,t+1} &= \frac{P_{i,t+1}S_{c_i,t+1} - P_{i,t}S_{c_i,t}}{P_{i,t}S_{c_i,t}} \\ &= \frac{P_{i,t}(1 + R_{i,t+1})\, S_{c_i,t}(1 + e_{c_i,t+1}) - P_{i,t}S_{c_i,t}}{P_{i,t}S_{c_i,t}} \\ &= R_{i,t+1} + e_{c_i,t+1} + R_{i,t+1}\, e_{c_i,t+1}. \end{aligned} \tag{3.1}$$

Let $\boldsymbol{x}_t \in \mathbb{R}^N$ be the vector of asset portfolio weights chosen at time t and held until $t{+}1$, where $x_{i,t}$ is the fraction of total portfolio value invested in asset i at time t. We impose the standard budget constraint $\mathbf{1}^\mathsf{T}\boldsymbol{x}_t = 1$.

Assuming no rebalancing between times t and $t{+}1$, the domestic-currency portfolio's unhedged return is given by

$$R^u_{\mathcal{P},t+1} = \sum_{i=1}^{N} x_{i,t}\, R^u_{i,t+1}.$$

3.1.2 Hedged Portfolio Returns

To manage currency risk, the investor can take positions in one-period currency forward contracts. For each foreign currency $c \in \mathcal{C}$, let $F_{c,t}$ be the forward rate quoted at time t for settlement at $t{+}1$, in units of domestic currency per unit of foreign currency c. We define the *forward premium* by

$$f_{c,t} \coloneqq \frac{F_{c,t} - S_{c,t}}{S_{c,t}}.$$

The investor hedges by selling foreign currency forward. Let $\phi_{c,t}$ denote the relative notional of the *short* forward position in currency c at time t, expressed as a fraction of total portfolio value, with the convention that $\phi_{c,t} > 0$ corresponds to hedging. Per unit of notional, the short position pays $f_{c,t} - e_{c,t+1}$ in domestic currency over the period. Adding these forward payoffs to the unhedged asset returns, the *hedged portfolio return* is

$$R^h_{\mathcal{P},t+1} = \underbrace{\sum_{i=1}^{N} x_{i,t}\, R^u_{i,t+1}}_{R^u_{\mathcal{P},t+1}} + \sum_{c=1}^{M} \phi_{c,t}\,(f_{c,t} - e_{c,t+1}). \tag{3.2}$$

3.1.3 Matrix Representation and Decomposition

To express the returns in a compact form suitable for optimization, we define the currency incidence matrix $\boldsymbol{C} \in \{0,1\}^{N\times M}$ by

$$C_{i,c} \coloneqq \begin{cases} 1, & \text{if asset } i \text{ is denominated in foreign currency } c, \\ 0, & \text{otherwise.} \end{cases}$$

Each row of $\boldsymbol{C}$ has at most one nonzero entry, and the rows corresponding to domestic assets are identically zero. This allows us to define the portfolio's *implicit foreign-currency exposure* as

$$\boldsymbol{w}_t \coloneqq \boldsymbol{C}^\mathsf{T}\boldsymbol{x}_t \in \mathbb{R}^M, \qquad w_{c,t} = \sum_{i:\, c_i=c} x_{i,t},$$

such that $w_{c,t}$ is the total portfolio weight invested in assets denominated in currency c. The implicit domestic exposure follows residually as $w_{0,t} = 1 - \mathbf{1}_M^\mathsf{T}\boldsymbol{w}_t$, since the asset weights sum to one.

We express returns in vector form $\boldsymbol{R}_{t+1} \in \mathbb{R}^N$, $\boldsymbol{e}_{t+1} \in \mathbb{R}^M$, and the forward premiums as $\boldsymbol{f}_t \in \mathbb{R}^M$. The product $\boldsymbol{C}\boldsymbol{e}_{t+1} \in \mathbb{R}^N$ aligns currency returns to assets, with ith component $e_{c_i,t+1}$ for foreign assets and 0 for domestic ones. Using (3.1), the vector of *unhedged domestic returns* across assets can be written as

$$\boldsymbol{R}^u_{t+1} = \boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}),$$

where $\odot$ denotes the Hadamard product.

Expressing the forward positions in vector form as $\boldsymbol{\phi}_t \in \mathbb{R}^M$, the hedged portfolio return (3.2) can be written as

$$R^h_{\mathcal{P},t+1} = \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + \boldsymbol{\phi}_t^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}). \tag{3.3}$$

Setting $\boldsymbol{\phi}_t = \boldsymbol{w}_t = \boldsymbol{C}^\mathsf{T}\boldsymbol{x}_t$ neutralizes the implicit currency exposure. Substituting this together with the unhedged return vector into (3.3), the *fully hedged portfolio return* is

$$\begin{aligned} R^{fh}_{\mathcal{P},t+1} &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + (\boldsymbol{C}^\mathsf{T}\boldsymbol{x}_t)^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) \\ &= \boldsymbol{x}_t^\mathsf{T}\Big[\boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1})\Big] + \boldsymbol{x}_t^\mathsf{T}\boldsymbol{C}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) \\ &= \boldsymbol{x}_t^\mathsf{T}\Big[\boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}) + \boldsymbol{C}\boldsymbol{f}_t - \boldsymbol{C}\boldsymbol{e}_{t+1}\Big] \\ &= \boldsymbol{x}_t^\mathsf{T}\Big[\boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{f}_t + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1})\Big] \\ &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^{fh}_{t+1}. \end{aligned}$$

The first-order currency terms $\pm\boldsymbol{x}_t^\mathsf{T}\boldsymbol{C}\boldsymbol{e}_{t+1}$ cancel, leaving the *fully hedged asset return vector*

$$\boldsymbol{R}^{fh}_{t+1} = \boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{f}_t + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}).$$

More generally, define the *net currency exposure* $\boldsymbol{\psi}_t \coloneqq \boldsymbol{w}_t - \boldsymbol{\phi}_t$. Imposing $\sum_{c=0}^M \phi_{c,t} = 1$ gives $\sum_{c=0}^M \psi_{c,t} = 0$, so the currency overlay is a zero-investment portfolio. Substituting $\boldsymbol{\phi}_t = \boldsymbol{w}_t - \boldsymbol{\psi}_t$ into (3.3) expresses the hedged return in terms of net exposures,

$$\begin{aligned} R^h_{\mathcal{P},t+1} &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + (\boldsymbol{w}_t - \boldsymbol{\psi}_t)^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) \\ &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^u_{t+1} + \boldsymbol{w}_t^\mathsf{T}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) + \boldsymbol{\psi}_t^\mathsf{T}(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t) \\ &= \boldsymbol{x}_t^\mathsf{T}\boldsymbol{R}^{fh}_{t+1} + \boldsymbol{\psi}_t^\mathsf{T}(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t), \end{aligned} \tag{3.4}$$

which we adopt throughout the following portfolio optimization formulations, as it makes the asset weights $\boldsymbol{x}_t$ and the net currency exposures $\boldsymbol{\psi}_t$ the explicit variables of the optimization.

3.2 Mean-Variance Optimization Formulation

3.2.1 The Mean-Variance Criterion

3.2.2 Regularized Mean-Variance Problem

3.2.3 Currency Overlay Optimization

3.2.4 Joint Optimization

3.3 Expected Shortfall Optimization Formulation

3.3.1 Expected Shortfall and the Rockafellar–Uryasev Representation

3.3.2 Regularized Expected Shortfall Problem

3.3.3 Currency Overlay Optimization

3.3.4 Joint Optimization

Chapter 4

Data and Implementation

In this chapter, we describe the dataset used for the empirical evaluation of the portfolio optimization strategies and the implementation of the backtesting framework. In the data section we detail the construction of the investable universe. We follow [18] to analyze asset and currency returns and volatillities and expand the analysis to different investable universe sizes, and look at the correlation dependence of asset and currency returns. In the implementation section we describe the modular structure of the backtesting framework, the workflow of the backtest, and the metrics used to evaluate potrfolio performance.

4.1 Data

The empirical evaluation of all portfolio optimization strategies in this thesis relies on a comprehensive dataset of international equities, foreign exchange spot rates, and forward exchange rates sourced from Refinitiv Datastream. The dataset spans from June 1990 to June 2023, covering several different market regimes, including the dot-com crash, the 2008 Global Financial Crisis, the European sovereign debt crisis, and the COVID-19 pandemic.

We construct the investable universe from 14 global equity markets: Australia, Canada, Denmark, Norway, Sweden, Hong Kong, Japan, New Zealand, Singapore, South Africa, Switzerland, the Eurozone, the United Kingdom, and the United States. The universe is rebuilt at the beginning of each year from the largest stocks by market capitalization at the time (a rolling-window approach that mitigates survivorship bias). At the start of each year we rank stocks within each market and restrict the following year's investments to that set. To study how the universe size affects the optimization approaches, we form three cross-sections, comprising the top 10, top 50, and top 100 stocks per market. Because the constituents turn over each year, each universe spans more distinct stocks over the full sample than are held on any single date, 407 for the top 10, 1705 for the top 50, and 3189 for the top 100, respectively.

The US dollar is the base currency throughout our analysis. For each stock we compute fully hedged returns, and for each currency the excess currency return, following the model-free return representation in Chapter 3.

Table 4.1 reports annualized average returns and volatilities for these fully hedged equity returns and excess currency returns, by economy. For each economy, the asset figures are the cross-sectional average of the daily returns of its individual stocks.

We can see that as the investable universe becomes larger, the average returns tend to fall. In the top 10 universe every economy has a positive average return, ranging from 2.76% (the United Kingdom) to 12.59% (the United States). In the top 50 universe, the range drops

Economies	Assets (%)						Currencies (%)	
	Top 10		Top 50		Top 100			
	Return	Vol.	Return	Vol.	Return	Vol.	Excess Return	Vol.
Australia	4.57	29.94	3.10	37.72	-1.88	58.00	1.31	11.59
Canada	9.10	32.89	6.58	40.24	1.86	59.24	-0.03	7.63
Denmark	8.35	34.05	3.04	39.00	-3.47	48.52	-0.08	9.68
Eurozone	5.09	32.24	5.05	34.01	3.25	35.46	-0.40	9.58
Hong Kong	4.42	37.59	-0.15	45.77	-3.48	52.59	-0.16	0.56
Japan	5.37	33.69	4.41	34.01	4.09	35.38	-2.07	10.47
New Zealand	2.88	29.93	-1.13	37.16	-4.56	41.49	2.48	11.36
Norway	3.17	43.30	-1.59	50.29	-2.61	53.26	-0.36	11.73
Singapore	3.32	35.43	-1.99	54.57	-4.84	62.12	0.18	5.35
South Africa	3.53	37.72	1.01	43.84	-4.39	53.94	1.22	15.18
Sweden	9.54	34.50	6.92	42.50	5.37	47.65	-1.16	11.50
Switzerland	6.38	29.04	5.67	30.32	3.97	33.65	-0.12	10.99
United Kingdom	2.76	31.50	2.63	34.71	1.29	36.22	0.12	9.42
United States	12.59	33.58	7.99	39.07	6.86	42.84	0.00	0.00

Table 4.1: Summary statistics of fully hedged equity returns and excess currency returns by economy. Asset statistics are reported for three investment universes (the top 10, 50, and 100 stocks per economy). Excess currency returns are defined as the currency return net of the forward premium. All numbers are reported as annualized percentages, from the perspective of a USD-based investor.

downward, from -1.99% (Singapore) to 7.99% (the United States), and in the top 100 universe the range goes from -4.84% (Singapore) to 6.86% (the United States), with many economies now having negative average returns. The trend reflects the firm size, the largest stocks post the strongest returns, and the smaller firms added pull the average down. Volatilities exhibit the opposite pattern, rising with universe size, as smaller, more volatile stocks join the investable universe. Switzerland is the least volatile market in every case, its volatility rises from 29.04% to 33.65%, and the upper end rises from 43.30% (Norway) in the top 10 to 62.12% (Singapore) in the top 100.

Excess currency returns range between -2.07% for the Japanese yen and 2.48% for the New Zealand dollar, and excess currency volatilities range from 0.56% for the Hong Kong dollar, whose peg to the US dollar keeps it stable, up to 15.18% for the South African rand.

To quantify the dependence between asset and currency returns, we compute Pearson correlations between economy-level fully hedged equity returns and excess currency returns. For each economy we form an equally weighted daily portfolio of the fully hedged returns of its constituent stocks, resetting the constituents at each annual universe reconstitution. Correlating each economy's series with each currency's excess return over the full sample gives a 14×13 correlation matrix. The US dollar is dropped from the currency axis, since its excess return is zero by construction. We repeat this for the three universe sizes. The resulting heatmaps are shown in Figure 4.1.

Across all three universes the correlations range from -0.2 to 0.4, and are mostly positive. The pattern is intuitive and confirms well known empirical findings. The strongest positive correlations belong to commodity and risk-sensitive currencies such as the AUD, CAD, NZD and ZAR, whose equity markets and currencies rise together in good times and fall together in bad [3, 10, 11]. The JPY and CHF are the only negatively correlated currencies with most other economies (ranging from -0.2 to -0.05), in line with their role as safe havens that strengthen when equities fall [15]. The EUR, DKK, and HKD are minimally correlated, reflecting the HKD's peg to the US dollar, the DKK's peg to the EUR. This split between

risky and safe-haven currencies is well documented [2, 15].

This is the dependence structure the overlay optimization approach discards. It sets asset weights first, using only equity information, and adds the currency hedge afterwards, whereas the joint model can act on it directly. The structure barely changes across universe sizes. This matters for the results that follow, a shift in the relative performance of the joint and overlay approaches in Chapter 5 should therefore be attributed to estimation error rather than a change in the underlying dependence structure.

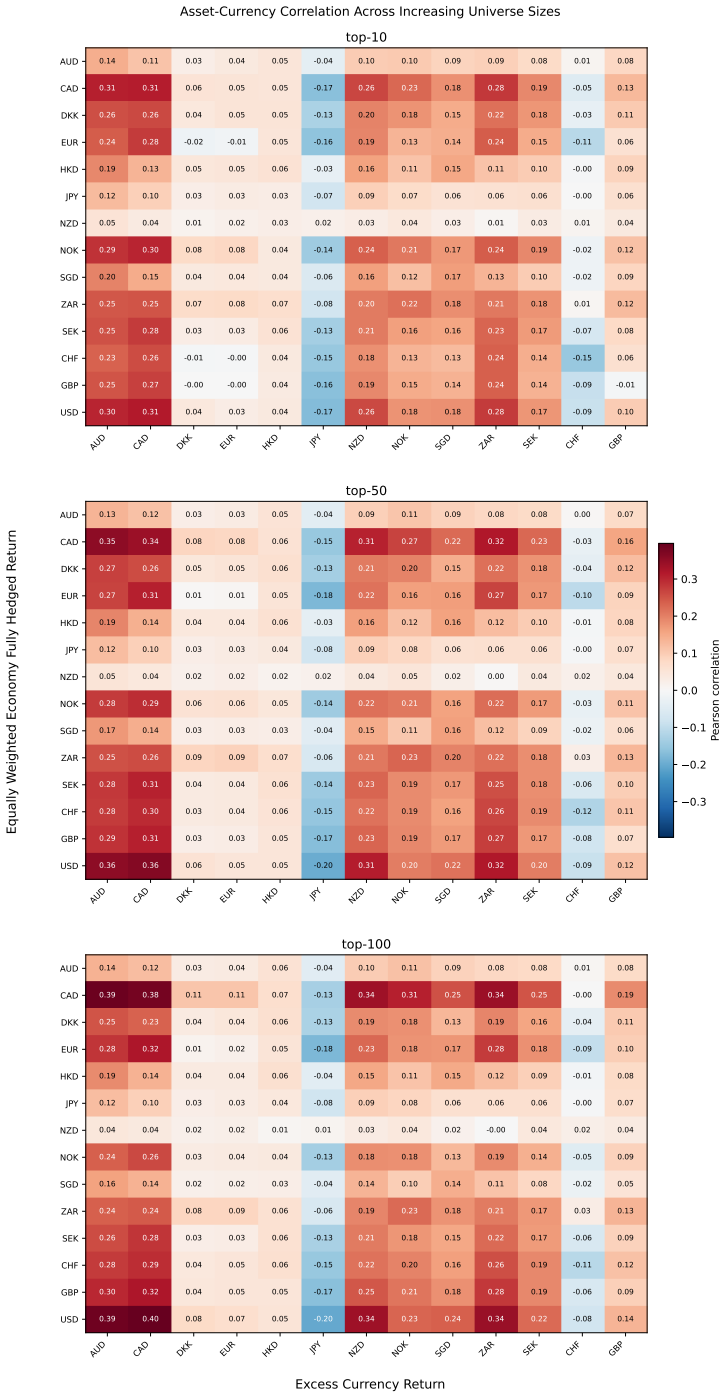


Figure 4.1: Pearson correlation between equally weighted economy fully hedged returns and excess currency returns, across the three investable universes. The US dollar is excluded from the currency axis, since it serves as the base currency and therefore its excess return is zero.

4.2 Implementation

This chapter describes the implementation of the walk-forward backtesting framework used throughout the thesis. The code is organised around a small number of clearly separated components: a data handler that prepares the investment universe and return series, a backtester that orchestrates the rolling evaluation, and a family of optimizers that implement the portfolio construction problem.[1] This modular structure keeps the experimental setup reproducible while allowing different objective functions and portfolio architectures to share the same evaluation pipeline.

4.2.1 Backtesting Framework

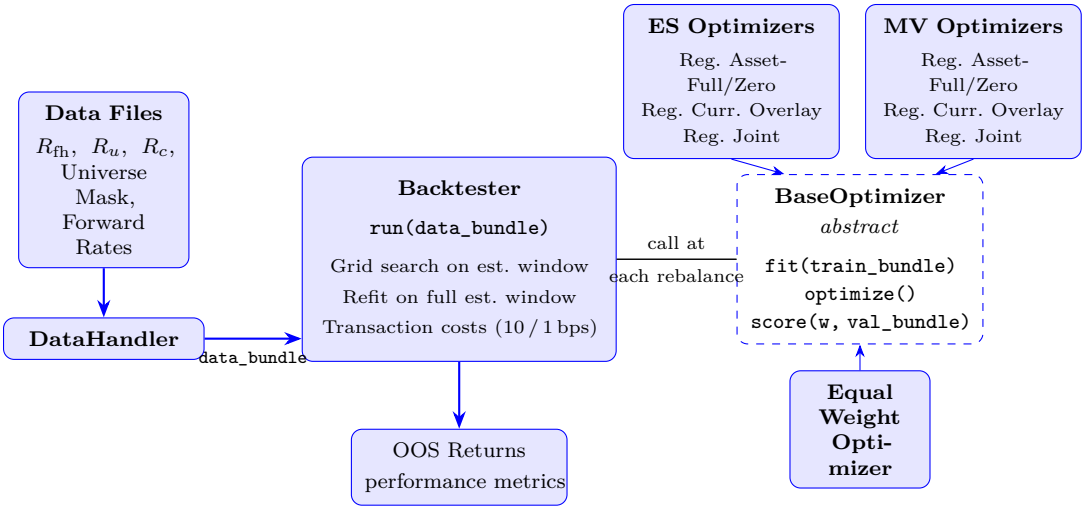


Figure 4.2: Schematic of the backtesting framework.

The figure highlights the main design principle of the implementation: the backtester is intentionally kept generic, while the portfolio logic lives in interchangeable optimizer classes. This makes it possible to compare mean-variance and expected shortfall formulations, asset-only and joint asset–currency strategies, and regularized and unregularized variants under the same walk-forward protocol.

4.2.2 Backtesting Workflow and Optimizer Structure

The framework is built around a strict separation of concerns. The `DataHandler` prepares the inputs for the backtest by loading precomputed return series, mapping each asset to its reporting currency, and building a daily universe mask from the investment universe file. The backtester receives these objects as a single `data_bundle` and does not need to know how they were generated.

The `Backtester` is responsible for the evaluation protocol. It determines rebalancing dates, constructs estimation and validation windows, performs hyperparameter selection when a grid is supplied, and records the resulting out-of-sample returns. All portfolio construction decisions are delegated to subclasses of `BaseOptimizer`, which all implement `fit(train_bundle)`, `optimize()`, and `score(weights, val_bundle)`.

[1]The full codebase for this project is publicly available on GitHub: https://github.com/anejrozman/Multi-Currency-Portfolio-Optimization-with-General-Risk-Measures.

The backtest follows a standard walk-forward procedure. At each rebalance date, the framework forms a rolling estimation window and a separate validation window, evaluates candidate hyperparameters, refits the best configuration on the full estimation sample, and then carries the portfolio into the next holding period. Rebalancing can be daily, weekly, monthly, quarterly, semi-annual, or annual. Before optimization, the code filters the investable universe to currently eligible assets and checks that each selected asset has a complete return history, which keeps missing-data handling outside the optimization problem. Window lengths are stored centrally in `backtest_config.py`; the default setup uses a ten-year estimation window and a one-year validation window.

4.2.3 Optimization, Costs, and Outputs

Two optimizer families are implemented: expected shortfall and mean-variance. Expected shortfall is formulated as a convex program with auxiliary variables, optional ℓ_1 and ℓ_2 regularization, and an ℓ_1-norm leverage cap. Mean-variance uses Bayes–Stein shrinkage for expected returns and a lightly regularized covariance matrix. Both families are available in asset-only and joint asset–currency variants. The asset-only strategies use either hedged returns (*assets-full*) or unhedged returns (*assets-zero*), while the joint strategies solve for asset weights and currency hedge weights together. The equal-weight benchmark follows the same interface but returns a uniform allocation.

All substantive optimizers are solved with CVXPY. When available, Gurobi is used as the default solver with tight tolerances; otherwise, the code falls back to another compatible solver. Transaction costs are applied at each rebalance using separate assumptions for equities and FX, namely 10 basis points and 1 basis point. Costs are computed with a log-linear turnover approximation, and the backtester tracks drifted weights between rebalances so that turnover reflects actual portfolio evolution. The framework outputs daily out-of-sample returns and summary statistics including annualized volatility, maximum drawdown, Sharpe ratio, Sortino ratio, and an expected shortfall ratio.

4.3 Numerical Implementation

4.4 Performance Evaluation Metrics

All backtested portfolios are evaluated on the same set of performance metrics, computed from the out-of-sample return series. These include the expeted shortfall at confidence level 80%, annualized volatility, maximum drawdown, Sharpe ratio, Sortino ratio, expected shortfall ratio and the average turnover for both the asset and currency legs.

To calculate the expected shortfall at confidence level α, we first compute the value-at-risk at that level, which is the $(1-\alpha)$-quantile of the return distribution. Then, the expected shortfall is the average of all returns that are less than or equal to this VaR threshold.[2]

The annualized volatility is computed as the standard deviation of daily returns multiplied by the square root of the number of trading days in a year (assumed to be 252).

The maximum drawdown is the largest decline in the cumulative return series.

The Sharpe ratio is the ratio of the average excess return to the standard deviation of returns, annualized by multiplying by the square root of 252. It is the standard measure of risk-adjusted

[2]The formal definitions of value at risk and expected shortfall at confidence level α are given in Section A.1.

return, and is given by

$$\mathrm{SR} = \frac{\bar{R} - R_f}{\sigma_R}\sqrt{252}.$$

The Sortino ratio is a variation of the Sharpe ratio that only considers downside volatility. It is calculated as the average excess return divided by the standard deviation of negative returns, annualized by multiplying by the square root of 252

$$\mathrm{SOR} = \frac{\bar{R} - R_f}{\sigma_{\mathrm{down}}}\sqrt{252}.$$

The expected shortfall ratio is the ratio of the average return to the expected shortfall at confidence level 80%, and is given by

$$\mathrm{ESR} = \frac{\bar{R}}{\mathrm{ES}_{0.8}}.$$

Finally, the average turnover is computed as the average absolute change in weights between rebalances, separately for the asset and currency legs. This metric captures the trading activity and associated transaction costs of the portfolio strategies.

Chapter 5

Empirical Analysis and Baseline Results

In this chapter, we present an empirical analysis of the multi-currency portfolio optimization problem using historical data. We apply the model-free return representation derived in chapter **??** to compute realized returns for the portfolio strategies derived in section **??**. We evalutate the performance of the various strategies on two investment universes, a large investment universe consisting of 50 biggest stocks in the 14 largest developed markets and a small investment universe consisting of 10 biggest stocks in the same markets. We compare the performance of the optimized portfolios to a benchmark equally weighted portfolio.

5.1 Backtest Results of Expected Shortfall Optimization Approaches

We can see in table 5.1 that the performance of the expected shortfall optimization approaches varies across different universe sizes. In the small universe, the optimized portfolios generally outperform the equal weight benchmark in terms of risk-adjusted returns, as measured by the Sharpe Ratio and Sortino Ratio. However, as we increase the universe size to medium and large, the performance of the optimized portfolios relative to the equal weight benchmark deteriorates, with some strategies even underperforming the benchmark in terms of risk-adjusted returns. This suggests that the benefits of optimization may diminish as the investment universe expands, potentially due to increased estimation error and model uncertainty in larger universes.

5.2 Backtest Results of Mean-Variance Optimization Approaches

5.3 The Effects of Universe Size on Portfolio Performance

Panel A: Small Investment Universe (top 10 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)	1.15			0.69		0.67	
Vol (%)	13.96			8.18		7.97	
MD (%)	54.76			29.31		28.31	
SR	0.12			0.08		0.08	
SOR	0.17			0.10		0.11	
ESR	0.09			0.06		0.06	
Turnover^a	0.29			1.28		1.26	
Turnover^c	0.92			2.39		2.98	
Panel B: Medium Investment Universe (top 50 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)	1.04			0.44		0.43	
Vol (%)	12.50			5.68		5.53	
MD (%)	55.74			38.59		38.84	
SR	0.32			0.56		0.54	
SOR	0.44			0.75		0.73	
ESR	0.24			0.445		0.43	
Turnover^a	0.27			1.78		1.78	
Turnover^c	0.92			0.97		1.67	
Panel C: Large Investment Universe (top 100 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)	0.97			0.34		0.35	
Vol (%)	11.67			4.57		4.55	
MD (%)	56.26			31.32		30.45	
SR	0.37			0.72		0.76	
SOR	0.49			0.95		1.01	
ESR	0.28			0.59		0.62	
Turnover^a	0.27			1.89		1.91	
Turnover^c	0.91			0.97		1.43	

Table 5.1: Performance metrics for Expected Shortfall optimization approaches on increasing universe sizes

Table 5.1 reports out-of-sample performance metrics for the international portfolio optimization under expected shortfall, with portfolios rebalanced quarterly. The realized out-of-sample statistics shown are Expected Shortfall (ES), volatility (Vol), maximum drawdown (MD), the Sharpe Ratio (SR), the Sortino Ratio (SOR), the Expected Shortfall Ratio (ESR), and average asset and currency turnover, denoted Turnover^a and Turnover^c, respectively. Each panel corresponds to a different investment universe: Panel A covers the small universe, made up of the 10 largest companies by market capitalization in each economy; Panel B the medium universe, comprising the 50 largest companies per economy; and Panel C the large universe, consisting of the 100 largest companies per economy. The sample period runs from June 1990 to June 2023, and the first out-of-sample portfolio is formed in June 2000. Details on the optimization approaches and performance metrics are provided in section **??**.

Panel A: Small Investment Universe (top 10 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)							
Vol (%)							
MD (%)							
SR							
SOR							
ESR							
Turnover^a							
Turnover^c							
Panel B: Medium Investment Universe (top 50 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)							
Vol (%)							
MD (%)							
SR							
SOR							
ESR							
Turnover^a							
Turnover^c							
Panel C: Large Investment Universe (top 100 stocks per economy)							
Metric	**Equal Weight**	**Assets-Full**	**Assets-Zero**	**Overlay**	**Overlay-Reg**	**Joint**	**Joint-Reg**
ES (%)							
Vol (%)							
MD (%)							
SR							
SOR							
ESR							
Turnover^a							
Turnover^c							

Table 5.2: Performance metrics for mean-variance optimization approaches on increasing universe sizes

Table 5.2 reports out-of-sample performance metrics for the international portfolio optimization under mean-variance, with portfolios rebalanced quarterly. The realized out-of-sample statistics shown are Expected Shortfall (ES), volatility (Vol), maximum drawdown (MD), the Sharpe Ratio (SR), the Sortino Ratio (SOR), the Expected Shortfall Ratio (ESR), and average asset and currency turnover, denoted Turnover^a and Turnover^c, respectively. Each panel corresponds to a different investment universe: Panel A covers the small universe, made up of the 10 largest companies by market capitalization in each economy; Panel B the medium universe, comprising the 50 largest companies per economy; and Panel C the large universe, consisting of the 100 largest companies per economy. The sample period runs from June 1990 to June 2023, and the first out-of-sample portfolio is formed in June 2000. Details on the optimization approaches and performance metrics are provided in section **??**.

Chapter 6

Convex Risk Measures

Chapter 7

End To End Portfolio Optimization

Chapter 8

Conclusion

8.1 Findings

8.2 Limitations and Future Research

Appendices

Appendix A

Optimization Derivation: Expected Shortfall

In this appendix, we derive the optimization problems used in the main text, building up from the Expected Shortfall objective to the implementable quadratic programs solved in the empirical analysis. In Section A.1 we follow [16] to derive the equivalence between direct minimization of Expected Shortfall and an auxiliary formulation suitable for linear programming. Following [18], in Section A.2 we embed this equivalence in a mean-ES objective, by replacing the conditional expectation with its empirical counterpart, and linearizing the terms to assemble the base unregularized portfolio optimization problem as a linear program. In Section A.3 we introduce three regularization techniques, an L_1 penalty, an L_2 penalty, and a leverage constraint, giving us the base regularized quadratic program for portfolio optimization. In Section A.4 we extend the framework to the multi-currency setting under the separate (currency overlay) approach, deriving the asset and the currency quadratic programs. Finally, in Section A.5 we derive the joint quadratic program formulation, in which asset and currency weights are optimized simultaneously through a single Expected Shortfall term that captures the combined return distribution.

A.1 Derivation of the Auxiliary Expected Shortfall Formulation

We derive the equivalence between the direct minimization of Expected Shortfall and the auxiliary formulation proposed by Rockafellar and Uryasev [16] that allows for formulating the portfolio optimization as a linear program.

We assume that the loss distribution is continuous, meaning the loss cumulative distribution function has no jumps and the probability density function $f(l)$ exists. Under these assumptions, Value-at-Risk (VaR) is uniquely defined [14].

Define the loss random variable at time $t+1$, given portfolio weights $\boldsymbol{x}_t$, as $L(\boldsymbol{x}_t) = -\boldsymbol{x}_t^T \boldsymbol{R}_{t+1}$. Let $f(l)$ denote the probability density function of the loss $L(\boldsymbol{x}_t)$. The VaR at confidence level β is defined as the β-quantile of the loss distribution

$$\mathrm{VaR}_\beta(\boldsymbol{x}_t) = \min\{\alpha : \mathbb{P}(L(\boldsymbol{x}_t) \leq \alpha) \geq \beta\}.$$

The Expected Shortfall at confidence level β is defined as the conditional expectation of losses exceeding VaR_β

$$\mathrm{ES}_\beta(\boldsymbol{x}_t) = \mathbb{E}[L(\boldsymbol{x}_t) \mid L(\boldsymbol{x}_t) \geq \mathrm{VaR}_\beta(\boldsymbol{x}_t)] = \frac{1}{1-\beta} \int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^{\infty} l f(l)\, dl.$$

We now consider the auxiliary function F_β defined as

$$F_\beta(\boldsymbol{x}_t, \alpha) = \alpha + \frac{1}{1-\beta}\mathbb{E}_t[(L(\boldsymbol{x}_t) - \alpha)^+].$$

We can rewrite the expectation term using the integral form

$$\mathbb{E}_t[(L(\boldsymbol{x}_t) - \alpha)^+] = \int_{-\infty}^{\infty} (l-\alpha)^+ f(l)\, dl = \int_{\alpha}^{\infty} (l-\alpha) f(l)\, dl.$$

Before minimizing $F_\beta(\boldsymbol{x}_t, \alpha)$ with respect to α, we verify convexity. Using Leibniz's rule, the first and second derivative of F_β with respect to α is

$$\begin{aligned}
\frac{\partial F_\beta}{\partial \alpha} &= 1 + \frac{1}{1-\beta}\frac{\partial}{\partial \alpha}\int_{\alpha}^{\infty} (l-\alpha) f(l)\, dl \\
&= 1 + \frac{1}{1-\beta}\left[-f(\alpha)\cdot 0 + \int_{\alpha}^{\infty} (-1) f(l)\, dl\right] \\
&= 1 - \frac{1}{1-\beta}\int_{\alpha}^{\infty} f(l)\, dl \\
\frac{\partial^2 F_\beta}{\partial \alpha^2} &= \frac{1}{1-\beta} f(\alpha).
\end{aligned}$$

Since the probability density function $f(\alpha) \geq 0$ and $\beta < 1$, the second derivative is non-negative, $\frac{\partial^2 F_\beta}{\partial \alpha^2} \geq 0$. Thus, $F_\beta(\boldsymbol{x}_t, \alpha)$ is convex in α, ensuring that the first-order condition is necessary and sufficient for a global minimum.

Differentiation of $F_\beta(\boldsymbol{x}_t, \alpha)$ with respect to α gives us the first-order condition for optimality

$$\begin{aligned}
\frac{\partial F_\beta}{\partial \alpha} &= 1 - \frac{1}{1-\beta}\int_{\alpha}^{\infty} f(l)\, dl \\
&= 1 - \frac{1}{1-\beta}\mathbb{P}(L(\boldsymbol{x}_t) \geq \alpha).
\end{aligned}$$

Setting the derivative to zero to find the optimal α^*

$$\begin{aligned}
1 - \frac{1}{1-\beta}\mathbb{P}(L(\boldsymbol{x}_t) \geq \alpha^*) &= 0 \\
\mathbb{P}(L(\boldsymbol{x}_t) \geq \alpha^*) &= 1-\beta \\
\mathbb{P}(L(\boldsymbol{x}_t) \leq \alpha^*) &= \beta.
\end{aligned}$$

This implies that the optimal α^* that minimizes the auxiliary function for a fixed $\boldsymbol{x}_t$ is exactly the Value-at-Risk at level β, so $\alpha^* = \mathrm{VaR}_\beta(\boldsymbol{x}_t)$.

Substituting $\alpha^* = \mathrm{VaR}_\beta(\boldsymbol{x}_t)$ back into $F_\beta(\boldsymbol{x}_t, \alpha)$ we obtain

$$\begin{aligned}
F_\beta(\boldsymbol{x}_t, \mathrm{VaR}_\beta(\boldsymbol{x}_t)) &= \mathrm{VaR}_\beta(\boldsymbol{x}_t) + \frac{1}{1-\beta}\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^{\infty} (l - \mathrm{VaR}_\beta(\boldsymbol{x}_t)) f(l)\, dl \\
&= \mathrm{VaR}_\beta(\boldsymbol{x}_t) + \frac{1}{1-\beta}\left(\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^{\infty} l f(l)\, dl - \mathrm{VaR}_\beta(\boldsymbol{x}_t)\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^{\infty} f(l)\, dl\right).
\end{aligned}$$

Recognizing that $\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^{\infty} f(l)\, dl = \mathbb{P}(L(\boldsymbol{x}_t) \geq \mathrm{VaR}_\beta(\boldsymbol{x}_t)) = 1-\beta$, we have

$$\begin{aligned}
F_\beta(\boldsymbol{x}_t, \mathrm{VaR}_\beta(\boldsymbol{x}_t)) &= \mathrm{VaR}_\beta(\boldsymbol{x}_t) + \frac{1}{1-\beta}\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^{\infty} l f(l)\, dl - \frac{1}{1-\beta}\mathrm{VaR}_\beta(\boldsymbol{x}_t)(1-\beta) \\
&= \mathrm{VaR}_\beta(\boldsymbol{x}_t) + \mathrm{ES}_\beta(\boldsymbol{x}_t) - \mathrm{VaR}_\beta(\boldsymbol{x}_t) \\
&= \mathrm{ES}_\beta(\boldsymbol{x}_t).
\end{aligned}$$

Thus, minimizing $\mathrm{ES}_\beta(\boldsymbol{x}_t)$ with respect to $\boldsymbol{x}_t$ is equivalent to minimizing $F_\beta(\boldsymbol{x}_t, \alpha)$ jointly with respect to $\boldsymbol{x}_t$ and α

$$\min_{\boldsymbol{x}_t} \mathrm{ES}_\beta(\boldsymbol{x}_t) = \min_{\boldsymbol{x}_t, \alpha} \left(\alpha + \frac{1}{1-\beta} \mathbb{E}_t[(-\boldsymbol{x}_t^T \boldsymbol{R}_{t+1} - \alpha)^+] \right). \tag{A.1}$$

Finally, we observe that $F_\beta(\boldsymbol{x}_t, \alpha)$ is jointly convex in $(\boldsymbol{x}_t, \alpha)$. The term α is linear (hence convex). The inner map $(\boldsymbol{x}_t, \alpha) \mapsto (-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)$ is affine. The outer function $z \mapsto z^+$ is convex and nondecreasing. The composition of a convex function with an affine map is convex [1]. Thus $(-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)^+$ is jointly convex in $(\boldsymbol{x}_t, \alpha)$. Taking the expectation preserves convexity.

Since the objective function is convex, any local minimum is global, ensuring the optimization problem can be solved reliably with standard convex optimization methods.

A.2 Linear Program Representation of Portfolio Optimization

The equivalence in (A.1) reduces Expected Shortfall minimization to the minimization of the jointly convex auxiliary function $F_\beta(\boldsymbol{x}_t, \alpha)$

$$\min_{\boldsymbol{x}_t} \mathrm{ES}_\beta(\boldsymbol{x}_t) = \min_{\boldsymbol{x}_t, \alpha} \left(\alpha + \frac{1}{1-\beta} \mathbb{E}_t[(-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)^+] \right).$$

This formulation is not yet appropriate for portfolio optimization, as it only optimizes for downside risk, without taking into account the expected return of the portfolio. It also depends on the unknown expectation $\mathbb{E}_t[\cdot]$. We address this by incorporating the expected return into the optimization objective and replacing the expectation with its empirical estimator.

Following [6] and [18], we trade off expected return against downside risk through a relative risk-aversion coefficient $\gamma > 0$

$$\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \left\{ -\boldsymbol{\mu}_t^\top \boldsymbol{x}_t + \frac{\gamma}{2} \mathrm{ES}_\beta(\boldsymbol{x}_t) \right\}, \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1, \tag{A.2}$$

where $\boldsymbol{\mu}_t$ denotes the asset return mean estimator and the budget constraint $\boldsymbol{x}_t^\top \mathbf{1}_N = 1$ enforces full investment. Substituting the auxiliary representation (A.1) into (A.2) gives

$$\boldsymbol{x}_t^*, \alpha^* = \arg\min_{\boldsymbol{x}_t, \alpha} \left\{ -\boldsymbol{\mu}_t^\top \boldsymbol{x}_t + \frac{\gamma}{2}\alpha + \frac{\gamma}{2(1-\beta)} \mathbb{E}_t[(-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)^+] \right\}.$$

Let $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_q \in \mathbb{R}^N$ denote q historical realizations of the joint return vector $\boldsymbol{R}_{t+1}$. We approximate the conditional expectation by the empirical average over this sample

$$\mathbb{E}_t[(-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)^+] \approx \frac{1}{q} \sum_{k=1}^{q} (-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+,$$

so that the portfolio optimization problem becomes

$$\boldsymbol{x}_t^*, \alpha^* = \arg\min_{\boldsymbol{x}_t, \alpha} \left\{ -\boldsymbol{\mu}_t^\top \boldsymbol{x}_t + \frac{\gamma}{2}\alpha + \frac{\gamma}{2q(1-\beta)} \sum_{k=1}^{q} (-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+ \right\}, \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1. \tag{A.3}$$

The summand $(-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+$ is convex but non-linear. We linearize it by introducing, for each observation k, an auxiliary variable $u_k \geq 0$ subject to

$$u_k \geq -\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha \qquad \Longleftrightarrow \qquad \boldsymbol{x}_t^\top \boldsymbol{y}_k + \alpha + u_k \geq 0. \tag{A.4}$$

Since the objective in (A.3) is strictly increasing in each u_k (its coefficient $\frac{\gamma}{2q(1-\beta)} > 0$) and minimized, every u_k is driven to its lower bound at the optimum, so $u_k^* = (-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+$. The slack constraints therefore reproduce the term $(-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+$ while keeping all relations linear.

By collecting the decision variables in the vector

$$\tilde{\boldsymbol{x}}_t = \begin{bmatrix} x_{1,t}, \dots, x_{N,t}, & \alpha, & u_1, \dots, u_q \end{bmatrix}^\top = \begin{bmatrix} \boldsymbol{x}_t^\top, & \alpha, & \boldsymbol{u}^\top \end{bmatrix}^\top \in \mathbb{R}^{(N+q+1)},$$

and constants in the vector

$$\boldsymbol{c} = \begin{bmatrix} -\boldsymbol{\mu}_t^\top, & \frac{\gamma}{2}, & \frac{\gamma}{2q(1-\beta)} \mathbf{1}_q^\top \end{bmatrix}^\top \in \mathbb{R}^{(N+q+1)},$$

the portfolio optimization problem (A.3) can be written as a linear program

$$\begin{aligned} \tilde{\boldsymbol{x}}_t^* = \arg\min_{\tilde{\boldsymbol{x}}_t} \quad & \boldsymbol{c}^\top \tilde{\boldsymbol{x}}_t \\ \text{s.t.} \quad & \boldsymbol{D}\tilde{\boldsymbol{x}}_t = \begin{bmatrix} \mathbf{1}_N^\top & \mathbf{0}_{q+1}^\top \end{bmatrix} \tilde{\boldsymbol{x}}_t = E = 1, \\ & \boldsymbol{A}\tilde{\boldsymbol{x}}_t = \begin{bmatrix} -\boldsymbol{R}^{\text{data}} & -\mathbf{1}_q & -\boldsymbol{I}_q \end{bmatrix} \tilde{\boldsymbol{x}}_t \le \boldsymbol{b} = \mathbf{0}_q, \\ & \alpha \ge 0, \quad u_k \ge 0 \quad \text{for } k = 1, \dots, q. \end{aligned} \tag{A.5}$$

Here $\boldsymbol{D} \in \mathbb{R}^{1\times(N+q+1)}$ encodes the budget constraint, while $\boldsymbol{A} \in \mathbb{R}^{q\times(N+q+1)}$ and $\boldsymbol{b} \in \mathbb{R}^q$ encode the q inequality constraints arising from the linearization (A.4). The matrix $\boldsymbol{R}^{\text{data}} \in \mathbb{R}^{q\times N}$ is comprised of sampled return vectors $\boldsymbol{y}_k^\top$ row by row, $\mathbf{0}_q$ is the q-dimensional zero vector, and $\boldsymbol{I}_q$ the q-dimensional identity matrix.

Problem (A.5) has a linear objective subject to one linear equality and q linear inequalities, and is thus a linear program solvable with standard methods. This formulation serves as the base unregularized portfolio optimization problem under Expected Shortfall. Regularization terms and additional constraints are added in Section A.3 of the appendix.

A.3 Regularization

The base problem (A.5) is known to be numerically unstable in high-dimensional settings, producing extreme and concentrated weights [4, 8]. We stabilize it with three regularization techniques. An L_1-norm penalty that promotes sparsity, an L_2-norm penalty that promotes shrinkage, and a leverage constraint that bounds gross exposure. Each is appended to (A.5) while preserving its linear or quadratic structure. The L_1 penalty and the leverage constraint share a common variable decomposition, which we introduce first.

A.3.1 L_1 Regularization

The L_1 penalty $\lambda_1 \|\boldsymbol{x}_t\|_1 = \lambda_1 \sum_{i=1}^N |x_{i,t}|$, with $\lambda_1 \ge 0$, is non-linear through the absolute value. Following [17], we linearize it by splitting each weight into its positive and negative parts via the auxiliary variables $\Delta_{i,t}^+ \ge 0$ and $\Delta_{i,t}^- \ge 0$

$$x_{i,t} = \Delta_{i,t}^+ - \Delta_{i,t}^-, \qquad |x_{i,t}| = \Delta_{i,t}^+ + \Delta_{i,t}^-. \tag{A.6}$$

The second identity holds at the optimum. Minimizing the L_1 penalty drives the parts to their smallest feasible values, so at most one of $\Delta_{i,t}^+$, $\Delta_{i,t}^-$ is non-zero, giving $\Delta_{i,t}^+ = \max(x_{i,t}, 0)$ and $\Delta_{i,t}^- = \max(-x_{i,t}, 0)$. The L_1 penalty is then linear in the auxiliary variables

$$\lambda_1 \|\boldsymbol{x}_t\|_1 = \lambda_1 \mathbf{1}_N^\top (\boldsymbol{\Delta}_t^+ + \boldsymbol{\Delta}_t^-).$$

We extend the decision vector to carry the two parts

$$\tilde{\boldsymbol{x}}_t = \left[\boldsymbol{x}_t^\top,\ \ \alpha,\ \ \boldsymbol{u}^\top,\ \ (\boldsymbol{\Delta}_t^-)^\top,\ \ (\boldsymbol{\Delta}_t^+)^\top\right]^\top \in \mathbb{R}^{3N+q+1},$$

and append the penalty coefficients to the cost vector

$$\tilde{\boldsymbol{c}} = \left[-\boldsymbol{\mu}_t^\top,\ \ \tfrac{\gamma}{2},\ \ \tfrac{\gamma}{2q(1-\beta)}\mathbf{1}_q^\top,\ \ \lambda_1\mathbf{1}_N^\top,\ \ \lambda_1\mathbf{1}_N^\top\right]^\top.$$

The decomposition (A.6) is enforced as N additional linear equality constraints, $\boldsymbol{x}_t+\boldsymbol{\Delta}_t^- -\boldsymbol{\Delta}_t^+ = \mathbf{0}_N$, alongside the original budget constraint. All relations remain linear, so the optimization problem is still a linear program after the addition of the L_1 penalty.

A.3.2 L_2 Regularization

The L_2 penalty $\lambda_2\|\boldsymbol{x}_t\|_2^2 = \lambda_2\boldsymbol{x}_t^\top\boldsymbol{x}_t$, with $\lambda_2 \geq 0$, is quadratic in the weights and is written as a quadratic form

$$\lambda_2\|\boldsymbol{x}_t\|_2^2 = \tfrac{1}{2}\,\boldsymbol{x}_t^\top\boldsymbol{Q}\,\boldsymbol{x}_t, \qquad \boldsymbol{Q} = 2\lambda_2\boldsymbol{I}_N.$$

Embedding this in the augmented vector $\tilde{\boldsymbol{x}}_t$, the penalty becomes $\frac{1}{2}\,\tilde{\boldsymbol{x}}_t^\top\tilde{\boldsymbol{Q}}\,\tilde{\boldsymbol{x}}_t$ with $\tilde{\boldsymbol{Q}}$ placing the block $2\lambda_2\boldsymbol{I}_N$ on the asset weights and zeros on the auxiliary variables

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} 2\lambda_2\boldsymbol{I}_N & \mathbf{0}_{N\times(2N+q+1)} \\ \mathbf{0}_{(2N+q+1)\times N} & \mathbf{0}_{(2N+q+1)\times(2N+q+1)} \end{bmatrix}.$$

Adding this term turns the L_1-augmented linear objective of the problem (A.5) into a quadratic objective $\tilde{\boldsymbol{c}}^\top\tilde{\boldsymbol{x}}_t + \frac{1}{2}\,\tilde{\boldsymbol{x}}_t^\top\tilde{\boldsymbol{Q}}\,\tilde{\boldsymbol{x}}_t$. Since $\tilde{\boldsymbol{Q}}$ is positive semi-definite, the objective remains convex, and the optimization problem becomes a quadratic program with linear constraints.

A.3.3 Leverage Constraint

The leverage constraint $\|\boldsymbol{x}_t\|_1 \leq \ell$, with $\ell \geq 1$, bounds the portfolio's gross exposure and guarantees a bounded feasible region even when $\lambda_1 = 0$. Reusing the decomposition (A.6), it is expressed as a single linear inequality

$$\|\boldsymbol{x}_t\|_1 = \mathbf{1}_N^\top(\boldsymbol{\Delta}_t^+ + \boldsymbol{\Delta}_t^-) \leq \ell. \tag{A.7}$$

The choice $\ell = 1$ rules out leverage entirely and is equivalent to a hard L_1 constraint. As a linear inequality, it preserves the quadratic program structure.

A.3.4 Quadratic Program Representation of the Regularized Portfolio Optimization

Adding all three regularization terms to the base problem (A.5), makes the portfolio optimization problem under Expected Shortfall a quadratic program with linear constraints

$$\begin{aligned}
\tilde{\boldsymbol{x}}_t^* &= \arg\min_{\tilde{\boldsymbol{x}}_t} \quad \tilde{\boldsymbol{c}}^\top\tilde{\boldsymbol{x}}_t + \tfrac{1}{2}\,\tilde{\boldsymbol{x}}_t^\top\tilde{\boldsymbol{Q}}\,\tilde{\boldsymbol{x}}_t, \\
\text{s.t.} \quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{x}}_t &= \begin{bmatrix} \mathbf{1}_N^\top & \mathbf{0}_{q+1}^\top & \mathbf{0}_N^\top & \mathbf{0}_N^\top \\ \boldsymbol{I}_N & \mathbf{0}_{N\times(q+1)} & \boldsymbol{I}_N & -\boldsymbol{I}_N \end{bmatrix}\tilde{\boldsymbol{x}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \mathbf{0}_N \end{bmatrix}, \\
\tilde{\boldsymbol{A}}\tilde{\boldsymbol{x}}_t &= \begin{bmatrix} -\boldsymbol{R}^{\text{data}} & -\mathbf{1}_q & -\boldsymbol{I}_q & \mathbf{0}_{q\times N} & \mathbf{0}_{q\times N} \\ \mathbf{0}_N^\top & 0 & \mathbf{0}_q^\top & \mathbf{1}_N^\top & \mathbf{1}_N^\top \end{bmatrix}\tilde{\boldsymbol{x}}_t \leq \boldsymbol{b} = \begin{bmatrix} \mathbf{0}_q \\ \ell \end{bmatrix}, \\
\alpha \geq 0, &\quad u_k \geq 0, \quad \Delta_{i,t}^- \geq 0, \quad \Delta_{i,t}^+ \geq 0 \quad \text{for } i = 1,\ldots,N,\ k = 1,\ldots,q.
\end{aligned} \tag{A.8}$$

The first row of $\tilde{\boldsymbol{D}}$ enforces the budget constraint and the remaining N rows enforce the weight decomposition (A.6). The first q rows of $\tilde{\boldsymbol{A}}$ encode the inequality constraints from the linearization (A.4), and are extended with zeros for the new variables, and the final row of $\tilde{\boldsymbol{A}}$ imposes the leverage constraint (A.7).

A.4 Separate (Currency Overlay) Optimization

In the separate (currency overlay) portfolio optimization approach, the asset and currency allocations are optimized in two sequential steps. First, the asset weights $\boldsymbol{x}_t \in \mathbb{R}^N$ are optimized on the fully hedged returns, then, conditional on the optimal $\boldsymbol{x}_t^*$, the currency exposures $\boldsymbol{\psi}_t \in \mathbb{R}^M$ are optimized. Let

$$\begin{aligned}\boldsymbol{\mu}^{fh} &= \mathbb{E}_t[\boldsymbol{R}_{t+1}^{fh}] \in \mathbb{R}^N,\\ \boldsymbol{\mu}^{c} &= \mathbb{E}_t[\boldsymbol{e}_{t+1} - \boldsymbol{f}_t] \in \mathbb{R}^M,\end{aligned}$$

denote the expected fully hedged asset return and the expected excess currency return respectively. The two otimization steps are then given by

$$\begin{aligned}\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \Big\{ &-\boldsymbol{x}_t^\top \boldsymbol{\mu}^{fh} + \tfrac{\gamma}{2}\mathrm{ES}_\beta\left(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}^{fh}\right) + \lambda_1^a\|\boldsymbol{x}_t\|_1 + \lambda_2^a\|\boldsymbol{x}_t\|_2^2\Big\}\\ &\text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1,\ \|\boldsymbol{x}_t\|_1 \le \ell,\end{aligned} \tag{A.9}$$

$$\begin{aligned}\boldsymbol{\psi}_t^* \mid \boldsymbol{x}_t^* = \arg\min_{\boldsymbol{\psi}_t} \Big\{ &-\boldsymbol{\psi}_t^\top \boldsymbol{\mu}^{c} + \tfrac{\gamma}{2}\,\mathrm{ES}_\beta\Big(\boldsymbol{x}_t^{*\top} \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)\Big)\\ &+ \lambda_1^c\|\boldsymbol{\psi}_t\|_1 + \lambda_2^c\|\boldsymbol{\psi}_t\|_2^2\Big\}.\end{aligned} \tag{A.10}$$

The regularization parameters are kept separate across assets $(\lambda_1^a, \lambda_2^a)$ and currencies $(\lambda_1^c, \lambda_2^c)$.

A.4.1 Asset Optimization

The asset optimization step (A.9) is identical to the regularized single-currency program (A.8). It carries a budget constraint, L_1 and L_2 penalty terms, and a leverage constraint. It is therefore obtained directly from (A.8) by substituting the matrix of historically sampled fully hedged returns $\boldsymbol{R}^{fh} \in \mathbb{R}^{q\times N}$ for $\boldsymbol{R}^{\text{data}}$, the mean $\boldsymbol{\mu}^{fh}$ for $\boldsymbol{\mu}_t$, and the asset-specific penalties λ_1^a, λ_2^a. With the augmented vector

$$\tilde{\boldsymbol{x}}_t = \Big[\boldsymbol{x}_t^\top,\ \ \alpha_x,\ \ \boldsymbol{u}_x^\top,\ \ (\boldsymbol{\Delta}_{x,t}^-)^\top,\ \ (\boldsymbol{\Delta}_{x,t}^+)^\top\Big]^\top \in \mathbb{R}^{3N+q+1},$$

the cost vector and penalty matrix are

$$\tilde{\boldsymbol{c}} = \Big[-(\boldsymbol{\mu}^{fh})^\top,\ \ \tfrac{\gamma}{2},\ \ \tfrac{\gamma}{2q(1-\beta)}\mathbf{1}_q^\top,\ \ \lambda_1^a\mathbf{1}_N^\top,\ \ \lambda_1^a\mathbf{1}_N^\top\Big]^\top,$$

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} 2\lambda_2^a \boldsymbol{I}_N & \mathbf{0}_{N\times(2N+q+1)} \\ \mathbf{0}_{(2N+q+1)\times N} & \mathbf{0}_{(2N+q+1)\times(2N+q+1)}\end{bmatrix},$$

and the equality, inequality, and sign constraints are exactly those of (A.8) with $\boldsymbol{R}^{fh}$ in place of $\boldsymbol{R}^{\text{data}}$. The step yields the optimal asset weights $\boldsymbol{x}_t^*$, which are then fixed in the currency overlay optimization.

A.4.2 Currency Overlay Optimization

Conditional on $\boldsymbol{x}_t^*$, the currency step (A.10) minimizes the Expected Shortfall of the combined return $\boldsymbol{x}_t^{*\top}\boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)$ over the currency exposures $\boldsymbol{\psi}_t$. Applying the same auxiliary representation and the empirical approximation as in Section A.2, we introduce the auxiliary variable α_ψ and slack variables $u_{k,\psi} \geq 0$. Since $\boldsymbol{x}_t^*$ is now fixed, the slack constraint becomes

$$u_{k,\psi} \geq -\boldsymbol{\psi}_t^\top(\boldsymbol{e}_k - \boldsymbol{f}_t) - \boldsymbol{x}_t^{*\top}\boldsymbol{R}_k^{fh} - \alpha_\psi \quad \Longleftrightarrow \quad -\boldsymbol{\psi}_t^\top(\boldsymbol{e}_k - \boldsymbol{f}_t) - \alpha_\psi - u_{k,\psi} \leq \boldsymbol{x}_t^{*\top}\boldsymbol{R}_k^{fh},$$

where $\boldsymbol{R}_k^{fh}$ is the k-th sampled fully hedged return vector and $\boldsymbol{e}_k$ the k-th sampled currency return vector. The fixed asset-portfolio returns therefore appear on the right-hand side. We stack the sampled excess currency returns $(\boldsymbol{e}_k - \boldsymbol{f}_t)^\top$ row by row into $\boldsymbol{R}^c \in \mathbb{R}^{q\times M}$, and collect the fixed asset-portfolio returns in $\boldsymbol{R}_P^{fh} = \boldsymbol{R}^{fh}\boldsymbol{x}_t^* \in \mathbb{R}^q$. We linearize the currency L_1 penalty with the decomposition $\psi_{c,t} = \Delta_{\psi,c,t}^+ - \Delta_{\psi,c,t}^-$ as in Section A.3.1, and form the augmented vector

$$\tilde{\boldsymbol{\psi}}_t = \left[\boldsymbol{\psi}_t^\top,\ \ \alpha_\psi,\ \ \boldsymbol{u}_\psi^\top,\ \ (\boldsymbol{\Delta}_{\psi,t}^-)^\top,\ \ (\boldsymbol{\Delta}_{\psi,t}^+)^\top\right]^\top \in \mathbb{R}^{3M+q+1},$$

The cost vector and penalty matrix are

$$\tilde{\boldsymbol{c}} = \left[-(\boldsymbol{\mu}^c)^\top,\ \ \tfrac{\gamma}{2},\ \ \tfrac{\gamma}{2q(1-\beta)}\mathbf{1}_q^\top,\ \ \lambda_1^c\mathbf{1}_M^\top,\ \ \lambda_1^c\mathbf{1}_M^\top\right]^\top,$$

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} 2\lambda_2^c\boldsymbol{I}_M & \mathbf{0}_{M\times(2M+q+1)} \\ \mathbf{0}_{(2M+q+1)\times M} & \mathbf{0}_{(2M+q+1)\times(2M+q+1)} \end{bmatrix}.$$

The currency overlay optimization is then the quadratic program

$$\begin{aligned}
\boldsymbol{\psi}_t^* &= \arg\min_{\tilde{\boldsymbol{\psi}}_t} \quad \tilde{\boldsymbol{c}}^\top\tilde{\boldsymbol{\psi}}_t + \tfrac{1}{2}\,\tilde{\boldsymbol{\psi}}_t^\top\tilde{\boldsymbol{Q}}\,\tilde{\boldsymbol{\psi}}_t, \\
\text{s.t.}\quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{\psi}}_t &= \left[\boldsymbol{I}_M \quad \mathbf{0}_{M\times(q+1)} \quad \boldsymbol{I}_M \quad -\boldsymbol{I}_M\right]\tilde{\boldsymbol{\psi}}_t = \tilde{\boldsymbol{E}} = \mathbf{0}_M, \\
\tilde{\boldsymbol{A}}\tilde{\boldsymbol{\psi}}_t &= \left[-\boldsymbol{R}^c \quad -\mathbf{1}_q \quad -\boldsymbol{I}_q \quad \mathbf{0}_{q\times M} \quad \mathbf{0}_{q\times M}\right]\tilde{\boldsymbol{\psi}}_t \leq \boldsymbol{b} = \boldsymbol{R}_P^{fh}, \\
&\alpha_\psi \geq 0, \quad u_{k,\psi} \geq 0, \quad \Delta_{\psi,c,t}^- \geq 0, \quad \Delta_{\psi,c,t}^+ \geq 0 \quad \text{for } c = 1,\dots,M,\ k = 1,\dots,q.
\end{aligned} \tag{A.11}$$

The currency optimization differs from the asset optimization (A.9) in the following ways. First, there is no budget constraint, the currency portfolio is a zero-investment portfolio, so the domestic exposure $\psi_{0,t}$ is inferred residually and the equality block $\tilde{\boldsymbol{D}}$ enforces only the weight decomposition, with right-hand side $\tilde{\boldsymbol{E}} = \mathbf{0}_M$ rather than $[1, \mathbf{0}_M^\top]^\top$. Second, the fixed asset leg shifts the inequality right-hand side from $\mathbf{0}_q$ to $\boldsymbol{R}_P^{fh} = \boldsymbol{R}^{fh}\boldsymbol{x}_t^*$. Third, no leverage constraint is imposed on the currency exposures through matrix $\tilde{\boldsymbol{A}}$. The L_1 and L_2 regularizing terms are as before, through the last two blocks of $\tilde{\boldsymbol{c}}$ and the main block of $\tilde{\boldsymbol{Q}}$, so the currency overlay remains a convex quadratic program.

A.5 Joint Optimization

In the joint portfolio optimization approach, asset and currency weights are optimized simultaneously rather than sequentially. We stack them into the combined decision vector $\boldsymbol{\theta}_t = [\boldsymbol{x}_t^\top, \boldsymbol{\psi}_t^\top]^\top \in \mathbb{R}^{N+M}$ and the combined return vector $\boldsymbol{R}_{t+1} = [\boldsymbol{R}_{t+1}^{fh\top}, (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)^\top]^\top \in \mathbb{R}^{N+M}$, with joint mean $\boldsymbol{\mu} = \mathbb{E}_t[\boldsymbol{R}_{t+1}] \in \mathbb{R}^{N+M}$. The joint optimization problem is

$$\begin{aligned}
\boldsymbol{\theta}_t^* = \arg\min_{\boldsymbol{\theta}_t} &\left\{-\boldsymbol{\theta}_t^\top\boldsymbol{\mu} + \tfrac{\gamma}{2}\mathrm{ES}_\beta\left(\boldsymbol{\theta}_t^\top\boldsymbol{R}_{t+1}\right) + \|\boldsymbol{S}_1\boldsymbol{\theta}_t\|_1 + \|\boldsymbol{S}_2\boldsymbol{\theta}_t\|_2^2\right\}, \\
&\text{s.t. } \boldsymbol{\theta}_t^\top\boldsymbol{q}_{N,M} = 1, \quad \|\boldsymbol{L}\boldsymbol{\theta}_t\|_1 \leq \ell.
\end{aligned} \tag{A.12}$$

The following matrices and vector adapt the problem to the joint optimization setting

$$\boldsymbol{q}_{N,M} = \begin{bmatrix} \mathbf{1}_N \\ \mathbf{0}_M \end{bmatrix}, \quad \boldsymbol{L} = \begin{bmatrix} \boldsymbol{I}_N & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \mathbf{0}_{M\times M} \end{bmatrix},$$

$$\boldsymbol{S}_1 = \begin{bmatrix} \lambda_1^a \boldsymbol{I}_N & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \lambda_1^c \boldsymbol{I}_M \end{bmatrix}, \quad \boldsymbol{S}_2 = \begin{bmatrix} \sqrt{\lambda_2^a}\, \boldsymbol{I}_N & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \sqrt{\lambda_2^c}\, \boldsymbol{I}_M \end{bmatrix}.$$

The vector $\boldsymbol{q}_{N,M}$ applies the budget constraint to the asset weights only, so that $\boldsymbol{\theta}_t^\top \boldsymbol{q}_{N,M} = \boldsymbol{x}_t^\top \mathbf{1}_N = 1$, leaving the currency weights as a zero-investment overlay. The matrix $\boldsymbol{L}$ applies the leverage constraint to the asset weights only, giving $\|\boldsymbol{L}\boldsymbol{\theta}_t\|_1 = \|\boldsymbol{x}_t\|_1$. The diagonal regularization matrices $\boldsymbol{S}_1$ and $\boldsymbol{S}_2$ allow the asset and currency penalties to differ within a single objective,

$$\|\boldsymbol{S}_1\boldsymbol{\theta}_t\|_1 = \lambda_1^a\|\boldsymbol{x}_t\|_1 + \lambda_1^c\|\boldsymbol{\psi}_t\|_1, \qquad \|\boldsymbol{S}_2\boldsymbol{\theta}_t\|_2^2 = \lambda_2^a\|\boldsymbol{x}_t\|_2^2 + \lambda_2^c\|\boldsymbol{\psi}_t\|_2^2.$$

Applying the auxiliary representation of Section A.1 and the empirical approximation of Section A.2 to the joint return $\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}$, we introduce the auxiliary variable α and slack variables $u_k \geq 0$ for $k = 1, \ldots, q$, and linearize the L_1 penalty by decomposing all $N + M$ components of $\boldsymbol{\theta}_t$ into positive and negative parts as in Section A.3.1. Collecting the asset and currency parts separately, we form the augmented vector

$$\tilde{\boldsymbol{\theta}}_t = \Big[\, \boldsymbol{x}_t^\top,\ \ \boldsymbol{\psi}_t^\top,\ \ \alpha,\ \ \boldsymbol{u}^\top,\ \ (\boldsymbol{\Delta}_{x,t}^-)^\top,\ \ (\boldsymbol{\Delta}_{\psi,t}^-)^\top,\ \ (\boldsymbol{\Delta}_{x,t}^+)^\top,\ \ (\boldsymbol{\Delta}_{\psi,t}^+)^\top \,\Big]^\top \in \mathbb{R}^{3N+3M+q+1}.$$

Let $\boldsymbol{R}^{\text{all}} = [\, \boldsymbol{R}^{fh} \ \ \boldsymbol{R}^c \,] \in \mathbb{R}^{q\times(N+M)}$ denote the matrix of jointly sampled asset and currency returns. The cost vector and penalty matrix are

$$\tilde{\boldsymbol{c}} = \Big[\, -\boldsymbol{\mu}^\top,\ \ \tfrac{\gamma}{2},\ \ \tfrac{\gamma}{2q(1-\beta)}\mathbf{1}_q^\top,\ \ \lambda_1^a\mathbf{1}_N^\top,\ \ \lambda_1^c\mathbf{1}_M^\top,\ \ \lambda_1^a\mathbf{1}_N^\top,\ \ \lambda_1^c\mathbf{1}_M^\top \,\Big]^\top,$$

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} 2\lambda_2^a\boldsymbol{I}_N & \mathbf{0}_{N\times M} & \mathbf{0}_{N\times(2N+2M+q+1)} \\ \mathbf{0}_{M\times N} & 2\lambda_2^c\boldsymbol{I}_M & \mathbf{0}_{M\times(2N+2M+q+1)} \\ \mathbf{0}_{(2N+2M+q+1)\times N} & \mathbf{0}_{(2N+2M+q+1)\times M} & \mathbf{0}_{(2N+2M+q+1)\times(2N+2M+q+1)} \end{bmatrix}.$$

The joint portfolio optimization problem is then the quadratic program

$$\begin{aligned} \tilde{\boldsymbol{\theta}}_t^* = \arg\min_{\tilde{\boldsymbol{\theta}}_t} \quad & \tilde{\boldsymbol{c}}^\top\tilde{\boldsymbol{\theta}}_t + \tfrac{1}{2}\,\tilde{\boldsymbol{\theta}}_t^\top\tilde{\boldsymbol{Q}}\,\tilde{\boldsymbol{\theta}}_t, \\ \text{s.t.} \quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{\theta}}_t = & \begin{bmatrix} \boldsymbol{q}_{N,M}^\top & \mathbf{0}_{q+1}^\top & \mathbf{0}_{N+M}^\top & \mathbf{0}_{N+M}^\top \\ \boldsymbol{I}_{N+M} & \mathbf{0}_{(N+M)\times(q+1)} & \boldsymbol{I}_{N+M} & -\boldsymbol{I}_{N+M} \end{bmatrix}\tilde{\boldsymbol{\theta}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \mathbf{0}_{N+M} \end{bmatrix}, \\ \tilde{\boldsymbol{A}}\tilde{\boldsymbol{\theta}}_t = & \begin{bmatrix} -\boldsymbol{R}^{\text{all}} & -\mathbf{1}_q & -\boldsymbol{I}_q & \mathbf{0}_{q\times 2N} & \mathbf{0}_{q\times 2M} \\ \mathbf{0}_{N+M}^\top & \mathbf{0}_{q+1}^\top & \boldsymbol{q}_{N,M}^\top & & \boldsymbol{q}_{N,M}^\top \end{bmatrix}\tilde{\boldsymbol{\theta}}_t \leq \boldsymbol{b} = \begin{bmatrix} \mathbf{0}_q \\ \ell \end{bmatrix}, \\ & \alpha \geq 0, \quad u_k \geq 0, \quad \text{for } k = 1, \ldots, q, \\ & \Delta_{x,i,t}^{\pm} \geq 0, \quad \Delta_{\psi,c,t}^{\pm} \geq 0, \quad \text{for } i = 1, \ldots, N,\ c = 1, \ldots, M. \end{aligned} \tag{A.13}$$

In $\tilde{\boldsymbol{D}}$, the first row enforces the asset budget constraint $\boldsymbol{x}_t^\top\mathbf{1}_N = 1$, and the remaining $N + M$ rows enforce the weight decomposition (A.6) for all components of $\boldsymbol{\theta}_t$. In $\tilde{\boldsymbol{A}}$, the first q rows encode the linearization inequalities (A.4) applied to the joint return matrix $\boldsymbol{R}^{\text{all}}$. The final row imposes the asset-only leverage constraint (A.7), giving $\mathbf{1}_N^\top(\boldsymbol{\Delta}_{x,t}^+ + \boldsymbol{\Delta}_{x,t}^-) = \|\boldsymbol{x}_t\|_1 \leq \ell$.

The joint formulation solves a single quadratic program over $\tilde{\boldsymbol{\theta}}_t$, capturing the interaction between asset and currency exposures through the joint Expected Shortfall term and the

off-diagonal return covariation in $\boldsymbol{R}^{\text{all}}$. This comes at the cost of a higher-dimensional problem relative to the two smaller programs of the separate approach.

All quadratic programs derived in this appendix share the same convex structure and are solvable with the same quadratic programming solver. The auxiliary representation of Expected Shortfall ensures this uniformity. Each extension of the program, from the base regularized single currency formulation (A.8), to the joint multi-currency program above (A.13), modifies only the return matrix and the constraint blocks that encode the budget, leverage, and weight decomposition, without changing the program structure.

Appendix B

Optimization Derivation: Mean-Variance

B.1 Quadratic Program Representation of the Mean-Variance Portfolio Optimization

B.2 Regularization

B.2.1 L_1 Regularization

B.2.2 L_2 Regularization

B.2.3 Leverage Constraint

B.2.4 Quadratic Program Representation of the Regularized Portfolio Optimization

B.3 Separate (Currency Overlay) Optimization

B.3.1 Asset Optimization

B.3.2 Currency Overlay Optimization

B.4 Joint Optimization

Appendix C

Transaction Costs

In this appendix, we detail the modeling of transaction costs in the context of international portfolio optimization. Following [19, 18, 9], and keeping the notation introduced in Chapter 3, we model transaction costs as proportional to the volume of traded assets and currencies, ignoring any fixed trading costs. The return of the hedged portfolio is expressed as the sum of unhedged asset returns and gains from forward currency positions

$$R^h_{P,t+1} = \sum_{i=1}^{N} x_{i,t} R^u_{i,t+1} + \sum_{c=1}^{M} \phi_{c,t}(f_{c,t} - e_{c,t+1}). \tag{3.2}$$

Let τ^{asset}_{t+1} and τ^{curr}_{t+1} denote the total transaction costs associated with asset and currency trades, respectively. The portfolio return net of transaction costs is then given by

$$R^{h,net}_{P,t+1} = R^h_{P,t+1} - \tau^{asset}_{t+1} - \tau^{curr}_{t+1}. \tag{C.1}$$

The total transaction costs are computed as the sum of the costs incurred from trading individual assets and currencies, weighted by the trading volumes

$$\tau^{asset}_{t+1} = \sum_{i=1}^{N} \tau^{asset}_{i,t+1} \left(x_{i,t+1} - x^{-}_{i,t+1} \right),$$

$$\tau^{curr}_{t+1} = \sum_{c=1}^{M} \tau^{curr}_{c,t+1} \phi_{c,t+1} + \sum_{c=1}^{M} \tau^{curr}_{c,t+1} \left(w_{c,t+1} - w^{-}_{c,t+1} \right),$$

where $\tau^{asset}_{i,t+1}$ and $\tau^{curr}_{c,t+1}$ are the per-unit transaction costs for asset i and currency c. The terms $x^{-}_{i,t+1}$ and $w^{-}_{c,t+1}$ denote the asset weights and implied currency weights immediately prior to rebalancing at time $t+1$ (i.e., the drifted weights).

Asset transaction costs are incurred only when rebalancing individual asset holdings. Currency transaction costs, however, arise from two sources, adjustments in forward hedge positions (assuming the forward contract maturity matches the rebalancing frequency, the old contract expires and a new contract of full notional size $\phi_{c,t+1}$ must be established at each rebalancing date) and currency exchanges required to fund purchases of assets denominated in different currencies ($w_{c,t+1} - w^{-}_{c,t+1}$). If the portfolio is rebalanced more frequently than the forward contract maturity, existing forward contracts are kept open and only incremental adjustments are traded. In this case, the forward transaction cost term is modeled as the absolute difference, replacing $\sum_{c=1}^{M} \tau^{curr}_{c,t+1} \phi_{c,t+1}$ with $\sum_{c=1}^{M} \tau^{curr}_{c,t+1} (\phi_{c,t+1} - \phi^{-}_{c,t+1})$, where $\phi^{-}_{c,t+1}$ represents the drifted hedge weight. Following [19], these costs are aggregated at the currency level, operating under

the assumption that proceeds from selling assets denominated in currency c can fund the purchase of other assets in the same currency without incurring currency conversion fees.

To define the individual per-unit costs $\tau^{asset}_{i,t+1}$ and $\tau^{curr}_{c,t+1}$, we consider an asset i denominated in currency c_i. Let $P^{ask}_{i,t}$, $P^{bid}_{i,t}$, and $P_{i,t}$ represent the ask, bid, and mid prices of the asset, and let $S^{ask}_{c_i,t}$, $S^{bid}_{c_i,t}$, and $S_{c_i,t}$ represent the corresponding currency prices. If asset i is purchased at time t and sold at time $t+1$, its realized log return is given by

$$\begin{aligned}
\log\left(\frac{P^{bid}_{i,t+1}S^{bid}_{c_i,t+1}}{P^{ask}_{i,t}S^{ask}_{c_i,t}}\right) &= \log\left(\frac{\left(P_{i,t+1}-\frac{\delta^{asset}_{i,t+1}}{2}\right)\left(S_{c_i,t+1}-\frac{\delta^{curr}_{c_i,t+1}}{2}\right)}{\left(P_{i,t}+\frac{\delta^{asset}_{i,t}}{2}\right)\left(S_{c_i,t}+\frac{\delta^{curr}_{c_i,t}}{2}\right)}\right) \\
&= \log\left(\frac{P_{i,t+1}\left(1-\frac{\delta^{asset}_{i,t+1}}{2P_{i,t+1}}\right)S_{c_i,t+1}\left(1-\frac{\delta^{curr}_{c_i,t+1}}{2S_{c_i,t+1}}\right)}{P_{i,t}\left(1+\frac{\delta^{asset}_{i,t}}{2P_{i,t}}\right)S_{c_i,t}\left(1+\frac{\delta^{curr}_{c_i,t}}{2S_{c_i,t}}\right)}\right) \\
&= \log\left(\frac{P_{i,t+1}S_{c_i,t+1}}{P_{i,t}S_{c_i,t}}\right)+\log\left(\frac{1-\frac{\delta^{asset}_{i,t+1}}{2P_{i,t+1}}}{1+\frac{\delta^{asset}_{i,t}}{2P_{i,t}}}\right)+\log\left(\frac{1-\frac{\delta^{curr}_{c_i,t+1}}{2S_{c_i,t+1}}}{1+\frac{\delta^{curr}_{c_i,t}}{2S_{c_i,t}}}\right) \\
&= \log\left(\frac{P_{i,t+1}S_{c_i,t+1}}{P_{i,t}S_{c_i,t}}\right)-\log\left(\frac{1+\frac{\delta^{asset}_{i,t}}{2P_{i,t}}}{1-\frac{\delta^{asset}_{i,t+1}}{2P_{i,t+1}}}\right)-\log\left(\frac{1+\frac{\delta^{curr}_{c_i,t}}{2S_{c_i,t}}}{1-\frac{\delta^{curr}_{c_i,t+1}}{2S_{c_i,t+1}}}\right),
\end{aligned}$$

where δ^{asset}_i and $\delta^{curr}_{c_i}$ are the bid-ask spreads. Comparing this to equation (C.1), the transaction cost is precisely the difference between the actual realized log return and mid-price returs for the asset and currency.

Defining half of the bid-ask spreads as

$$\chi^{asset}_{i,t} = \frac{\overbrace{P^{ask}_{i,t}-P^{bid}_{i,t}}^{\delta^{asset}_i}}{2P_{i,t}} \quad \text{and} \quad \chi^{curr}_{c,t} = \frac{\overbrace{S^{ask}_{c,t}-S^{bid}_{c,t}}^{\delta^{curr}_c}}{2S_{c,t}},$$

the per-unit transaction costs for trading asset i and currency c become

$$\tau^{asset}_{i,t+1} = \begin{cases} \log\left(\frac{1+\chi^{asset}_{i,t+1}}{1-\chi^{asset}_{i,t}}\right) & \text{if } x_{i,t+1} > x^{-}_{i,t+1}, \\ \log\left(\frac{1-\chi^{asset}_{i,t}}{1+\chi^{asset}_{i,t+1}}\right) & \text{otherwise,} \end{cases}$$

and

$$\tau^{curr}_{c,t+1} = \begin{cases} \log\left(\frac{1+\chi^{curr}_{c,t+1}}{1-\chi^{curr}_{c,t}}\right) & \text{if } \phi_{c,t+1} > 0 \text{ or } w_{c,t+1} > w^{-}_{c,t+1}, \\ \log\left(\frac{1-\chi^{curr}_{c,t}}{1+\chi^{curr}_{c,t+1}}\right) & \text{otherwise.} \end{cases}$$

For our empirical analysis, we adopt the simplifying assumption that proportional bid-ask spreads remain constant both over time and across the cross-sections of both assets and currencies. We set the half-spreads to $\chi^{asset} = 0.001$ (10 bps) for all equities and $\chi^{curr} = 0.0001$ (1 bp) for all currencies, consistent with empirical finidngs in [7, 13, 12, 5]. Under these assumptions, the transaction costs simplify to constant values of $\tau^{asset} \approx 0.002$ and $\tau^{curr} \approx 0.0002$.

Appendix D

Hyperparameter Tuning and Additional Results

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[1] For further information please consult the ETH Zurich websites, e.g. https://ethz.ch/en/the-eth-zurich/education/ai-in-education.html and https://library.ethz.ch/en/researching-and-publishing/scientific-writing-at-eth-zurich.html (subject to change).